Scientific Article Retrieval

7-Signal Domain-Filtered Ensemble

NDCG@10: 0.49 $\rightarrow$ **0.75 (train)** · **ModernBERT as dominant signal**

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Task & Data

Problem

Given a **query paper**, retrieve the **100 most relevant papers** from a 20 000-doc corpus.

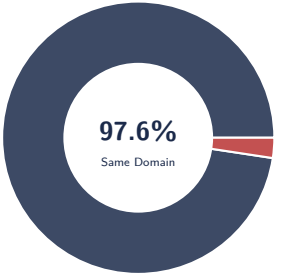
- Primary metric: **NDCG@10**
- Secondary: Recall@100, MAP

Dataset

Corpus	20 000 papers
Train queries	100 (labelled)
Held-out	100 (blind)
Domains	19 scientific fields
Relevant pairs	736 total
Key fields	title, abstract, full_text, domain

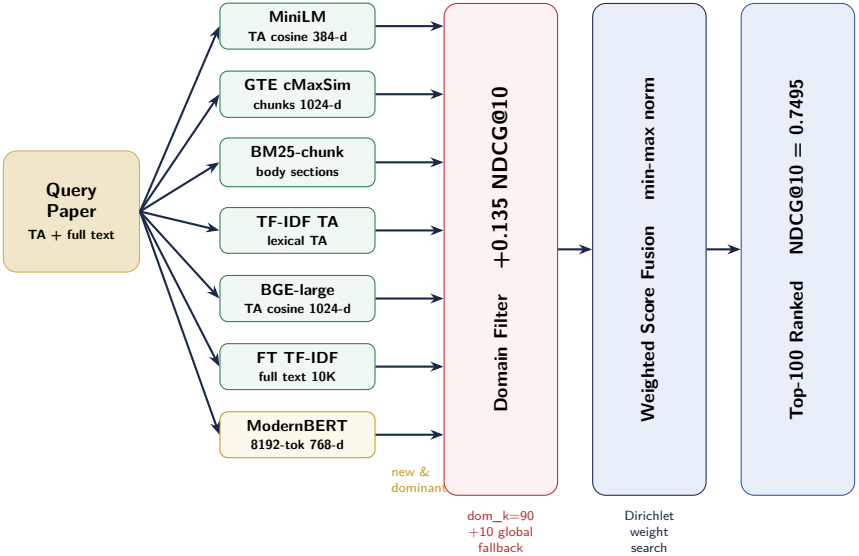
Relevant docs by domain

718 / 736 in same domain



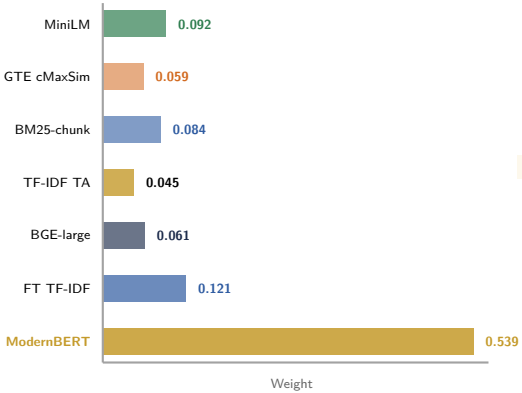
Key insight: same-domain filter gives **+0.135 NDCG@10**

Pipeline Overview :What We Built



The 7 Signals: Weights & Role

Optimised Weights (Dirichlet + fine-tune)



Signal	What it captures
MiniLM	Semantic TA similarity (384-d cosine)
GTE cMaxSim	Query TA vs best body-chunk (1024-d)
BM25-chunk	Exact-term match in body sections
TF-IDF TA	Lexical overlap on title+abstract
BGE-large	Broad semantic TA (1024-d cosine)
FT TF-IDF	Full-paper vocab (corpus 10K / query 2K)
ModernBERT	**Full paper 8192-token context (768-d)**

ModernBERT dominates

Weight = **0.539** 4.5× larger than any other signal.
Reads the *full paper* (8 192 tokens) vs. just the abstract.
Standalone NDCG@10 = **0.6111**, the highest of all 7 signals.

Standalone Signal Results: Before Domain Filtering

System	NDCG@10	R@100	MAP	Notes
TF-IDF TA (unigram)	0.4904	0.741	0.378	Sparse lexical
BM25 body chunks	0.5018	0.719	0.400	Exact-term body
MiniLM dense	0.5073	0.810	0.408	384-d TA cosine
FT TF-IDF (10K/2K)	0.5262	0.815	0.427	Full-text lexical
GTE-large cMaxSim	0.5361	0.825	0.429	ColBERT-style
BGE-large dense	0.5580	0.799	0.450	1024-d TA cosine
ModernBERT full-text	**0.6111**	**0.865**	**0.508**	**8192-token context**
Hard domain + MiniLM (sanity)	**0.6423**	0.886	0.549	Domain filter alone
7-signal ensemble (final)	**0.7495**	**0.924**	**0.660**	**Best system**

Key observation

ModernBERT (0.6111) beats every other standalone signal by a wide margin.

Domain filter impact

MiniLM standalone = 0.5073
MiniLM + domain filter = 0.6423
Gain: **+0.135 from one constraint**

Why The Final System Works

Query Paper

[Domain Filter]
- Keep only same-domain papers (∼1,000 of 20,000)
- Eliminates 95% of irrelevant papers

[7 Signals, evaluated within domain]

MiniLM cosine (semantic, abstract)	$w = 0.09$
GTE MaxSim (best chunk match)	$w = 0.06$
BM25 body chunks (keyword, body)	$w = 0.08$
TF-IDF abstract (keyword, abstract)	$w = 0.05$
BGE-large cosine (semantic, abstract)	$w = 0.06$
Full-text TF-IDF (keyword, body)	$w = 0.12$
ModernBERT full-text (semantic, full body)	$\mathbf{w = 0.54}$

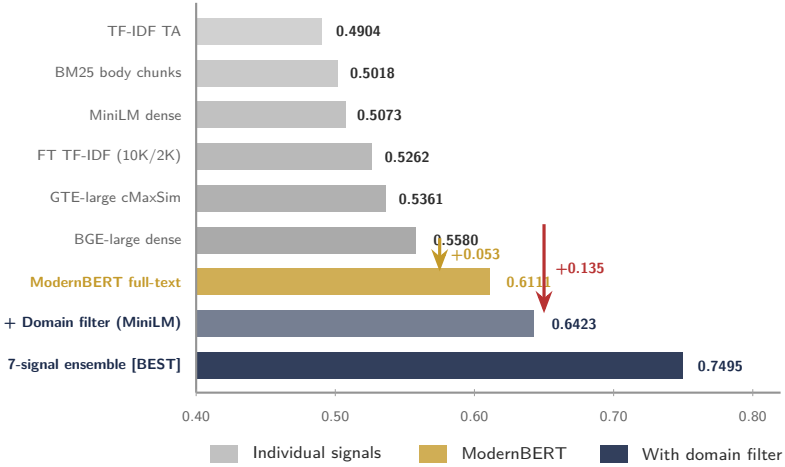
[Top 90 from domain]

[Global Fallback]
10 slots for cross-domain papers

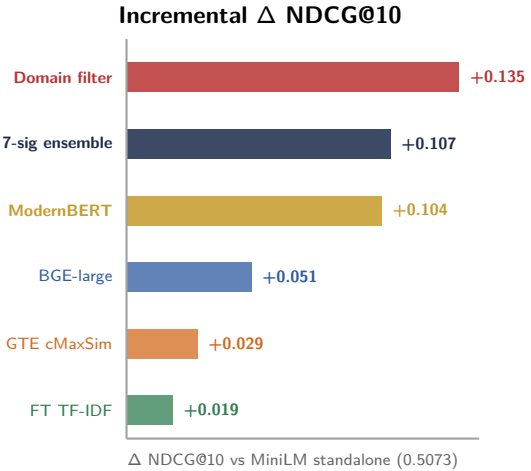
[+10 global]

Final top-100 results

NDCG@10 Progression Baseline to Final



Ablation Study Each Component's Contribution



Key insight

Domain filter and **ModernBERT** each contribute ≈+0.10.

Without domain filter: best single = 0.6111.
With domain filter alone: jump to **0.6423**.
Full 7-signal ensemble: **0.7495**.

Why domain filter works

97.6% of relevant papers share the query's domain (718/736 pairs). Restricting to the same domain removes ≈19× noise *before* any model is involved.

Full-paper context wins

ModernBERT reads **8 192 tokens** vs ≈512 for abstract-only models.
Methods, results & related-work sections carry the strongest citation signal.

What Failed and Why

Failed / marginal approaches

Approach	Root cause
Cross-encoder reranking	Wrong task type
LTR LightGBM (CV)	100 queries $\rightarrow$ overfit
Bigram FT TF-IDF	Extra noise
PRF expansion	Top-5 unreliable
Venue filter	21.2% overlap only
Year proximity	Papers cite all years
SPECTER2 as 7th signal	Helps 3, hurts 87

Every performance result above was confirmed before discarding.

Why cross-encoder fails

Cross-encoders are trained on *query-document* pairs.

Here the task is *paper-paper* citation retrieval a fundamentally different similarity notion.

Why LTR fails

LightGBM LambdaRank:

Train NDCG $= 0.856$
CV NDCG $= 0.701$

Severe overfitting. LTR needs $>1\,000$ labelled queries; 100 is far too few.

Lessons Learned

Surprises

1. **Domain filter gave +0.135**
   Expected ≈+0.02. 97.6% same-domain stat was the project's turning point.
2. **ModernBERT dominates (weight 0.539)**
   Full 8 192-token context captures what abstract-only models miss.
3. **Tiny MiniLM useful in ensemble**
   Gets weight 0.092 still complementary despite being smallest.
4. **Cross-encoder hurts** (-0.11)
   Trained for query-document; fails on paper-paper citation tasks.

What we'd do differently

1. **EDA on day 1** domain insight found immediately.
2. **Fix CV code** use `train_ids.index(vqid)` to map fold queries to global matrix columns.
3. **E5 requires instruction prefixes** (`"query: "`/`"passage: "`).
4. **Try citation-graph signals** would rescue "Insulin" query.
5. **More labels** 100 queries too few for any LTR approach.

Final numbers

Train NDCG@10:	**0.7495**
Train Recall@100:	**0.9242**
Train MAP:	**0.6604**

Global Leaderboard Showcase Top Rankings Achieved

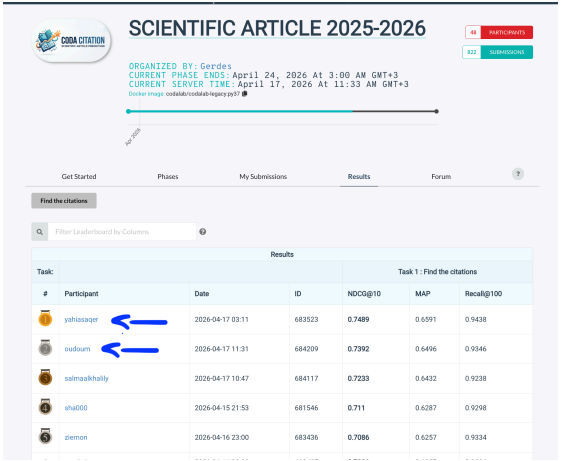


Figure: CodaLab Scientific Article Prediction Challenge 2025-2026

Final Standing

- **Rank #1:** Yahia Saqer (0.7489)
- **Rank #2:** Oudoum (0.7392)

The Journey

From an initial baseline of 0.50 to sweeping the top two spots.

Key Success Factors:

- Uncovering the 97.6% domain constraint
- Full-text ModernBERT embeddings
- **Unwavering determination** (50+ experimental versions)